Beyond In-Domain Discrimination: A Multi-Source Study of Identity-Learning Objectives for Cross-Domain Person Re-Identification

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Abstract. Person re-identification (Re-ID) models are often developed and evaluated within a single benchmark family, so their performance on familiar datasets may not predict transfer to a new camera network. Multi-source training broadens capture conditions but also exposes models to source-dependent annotation quality and substantial within-identity variation. We ask whether the identity-learning objective affects generalization to unseen domains. Using the FastReID Strong Baseline with a ResNet-50-IBN backbone, we compare eight candidate configurations: four identity-learning heads (linear softmax, ArcFace, Sub-center ArcFace, and SphereFace2), each trained with and without triplet loss. Each configuration is trained with three random seeds and evaluated on four unseen public benchmarks. ArcFace without triplet achieves the highest four-benchmark average among the eight candidates (44.8% mAP and 61.9% Rank-1), exceeding both its triplet variant (44.3% and 60.7%) and the FastReID default on the Strong Baseline (40.6% and 57.9%), with the largest gain on MSMT17 (+8.0 mAP). Triplet supervision helps the weaker heads but slightly hurts ArcFace and linear softmax, so its effect depends on the head it is paired with. The same objective does not transfer uniformly. It raises average mAP by 2.6 points on ConvNeXt-S but lowers it by 1.6 on OSNet, so the best result in the study is the Strong Baseline with ArcFace rather than the strongest backbone. Because the identity classifier is discarded at retrieval time, that gain has no inference cost. Under the default objective, the compact OSNet (43.6% and 60.8%) still beats the Strong Baseline with 12× fewer deployed parameters, while ConvNeXt-S is the weakest architecture under both objectives.

Keywords: Person re-identification · Identity-learning objectives · Domain generalization · Multi-source training · Angular-margin losses

1 Introduction

Person re-identification (Re-ID) aims to retrieve images of the same individual across non-overlapping camera views. Despite advances in deep representation learning, the

task remains challenging due to variations in viewpoint, illumination, occlusion, resolution, and clothing. Convolutional architectures have established effective and reproducible Re-ID baselines through learned identity representations and metric-learning objectives [**?**,**?**,**?**]. Compact designs such as OSNet further demonstrate their relevance to resource-constrained applications [**?**]. Although Transformer-based and vision-language approaches have advanced benchmark performance [**?**,**?**,**?**], CNN-based pipelines remain attractive when computational resources and model complexity are practical constraints.

An important challenge in deploying Re-ID systems is generalization to camera environments not represented during training. Differences in acquisition conditions and appearance distributions can cause substantial domain shifts, making strong in-domain performance an unreliable indicator of performance on unseen camera networks [**?**,**?**,**?**,**?**]. Multi-source training can improve exposure to diverse conditions, but combining heterogeneous datasets also introduces differences in annotation quality, sampling distributions, and within-identity appearance variation, particularly in datasets containing clothing changes, occlusion, and video-derived observations [**?**,**?**].

Beyond the training data and feature extractor, the identity-learning objective determines how the embedding space is structured. Classification objectives encourage identity discrimination, while metric-learning objectives such as triplet loss constrain relative distances between samples [**?**]. Alternative classification heads introduce different assumptions about class separation and intra-class variation, including angular margins, multiple class centers, and independent binary classification [**?**,**?**,**?**]. However, the relative behavior of these objectives under heterogeneous multi-source training and evaluation across unseen domains remains insufficiently characterized. This motivates a controlled investigation in which objective selection is examined independently of the feature-extraction architecture.

Accordingly, we first compare eight objective configurations using the FastReID Strong Baseline (SBS): four identity-learning heads, each evaluated with and without triplet loss, under a common training protocol and across three random seeds. The FastReID Circle-softmax-plus-triplet default is excluded from this loss sweep and serves instead as a reference. The best candidate of the sweep, ArcFace without triplet, is then compared with that default across the SBS, OSNet, and ConvNeXt-S. All models are evaluated on four public Re-ID benchmarks that are never used for training or checkpoint selection. Since the identity-classification head is discarded during retrieval inference, changing the training objective adds no classification-head computations at inference.

Contributions. This work makes three contributions. First, it presents a controlled, three-seed comparison of eight identity-learning objective configurations, covering four heads with and without triplet supervision under heterogeneous multi-source training. ArcFace alone improves the average held-out mAP of the Strong Baseline by 4.2 points over its default objective, and adding triplet loss never helps it. Second, it tests whether that objective transfers to OSNet and ConvNeXt-S. The gain is smaller on ConvNeXt-S and negative on OSNet, so a loss chosen on one backbone is not a default for another. The comparison also relates retrieval accuracy to parameter count, compute, and measured latency on a laptop GPU and CPU. Third, it establishes a harmonized nine-source Re-ID

training protocol with a common metadata representation, domain-balanced sampling, composite internal validation, and strictly held-out evaluation.

2 Related Work

Domain-generalizable person re-identification. Domain generalization in person Re-ID aims to learn identity representations from labeled source domains that remain effective on unseen camera networks without target-domain data. Prior approaches address this challenge through style normalization, feature-statistics mixing, and multi-source training strategies. Style Normalization and Restitution (SNR) suppresses style variations through instance normalization while restoring identity-relevant information [**?**]; MixStyle mixes feature statistics across training samples to synthesize style variation [**?**]; and Memory-based Multi-Source Meta-Learning (M^3L) simulates unseen-domain shifts with meta-learning, a memory-based identification loss, and meta-batch normalization [**?**]. Lee et al.'s 2025 survey reviews these and related domain-generalization strategies [**?**]. Whereas these methods modify representations or training procedures, our work isolates the identity-learning objective under a shared multi-source protocol.

Identity-learning objectives. Person Re-ID commonly employs identity classification and metric-learning objectives, either independently or jointly. Triplet loss encourages relative distance separation between samples of the same and different identities [**?**]. Angular-margin approaches, including SphereFace and ArcFace, introduce geometric constraints to improve inter-class discrimination [**?**,**?**]. Sub-center ArcFace extends this formulation by representing each identity with multiple centers, providing greater flexibility for intra-class variation and noisy labels [**?**]. SphereFace2 adopts an independent binary-classification formulation motivated by open-set recognition [**?**], whereas Circle-based objectives adapt the weighting of similarity relationships during optimization [**?**,**?**]. Recent work has also examined alternative loss combinations for cross-domain Re-ID. AATransID, for instance, evaluates combinations of ArcFace, Circle, CosFace, and additive-margin softmax with triplet supervision, reporting its strongest results with ArcFace and triplet loss [**?**]. Unlike this Transformer-based approach, our study compares a broader set of objective configurations within a common CNN-based training framework and evaluates their behavior across multiple held-out domains.

Convolutional architectures for person Re-ID. Convolutional architectures remain an established foundation for practical person Re-ID. Strong baselines combine classification supervision, metric learning, and batch-normalization necks to produce discriminative retrieval embeddings [**?**,**?**,**?**]. FastReID provides a unified framework for implementing and evaluating these components [**?**], while OSNet introduces omni-scale feature learning to capture discriminative information at different spatial scales with an efficient architecture [**?**]. Transformer-based alternatives, including TransReID and CLIP-ReID, have further expanded the architectural landscape through attention-based and vision-language representations [**?**,**?**]. Rather than introducing a new feature extractor, our work uses the FastReID Strong Baseline for the primary objective comparison and subsequently evaluates the selected formulation on OSNet and ConvNeXt-S to examine its transfer across different CNN architectures.

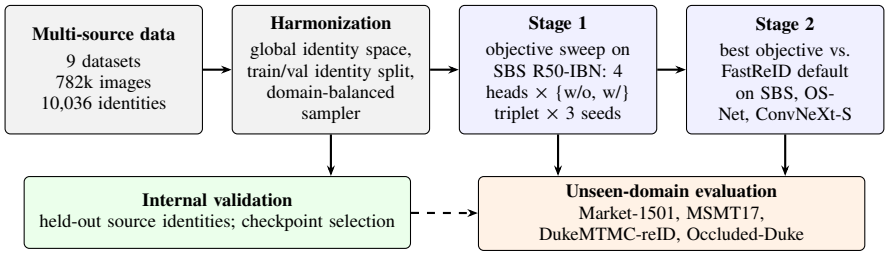


Fig. 1. Overview of the experimental framework. Nine harmonized sources are used for training and checkpoint selection; four public benchmarks are reserved exclusively for evaluation. Stage 1 compares eight objective configurations on a fixed architecture, and Stage 2 tests whether the best of them transfers to other backbones.

Table 1. Composition of the multi-source training dataset.

Dataset	Images
CUHK03-NP [?,?]	5,892
LaST [?]	71,248
Occluded-PoseTrack-ReID [?]	14,266
DeepChange [?]	60,916
LPW [?]	338,952
ORBench RGB [?]	21,564
PRID2011 [?]	31,690
RPIfield [?]	179,996
IUST PersonReID [?]	57,540
Total	**782,064**

3 Datasets

Three data partitions are used in this study: nine heterogeneous sources assembled for training and validation, and four public benchmarks reserved exclusively for unseen-domain evaluation. The overall experimental pipeline is illustrated in Fig. 1.

3.1 Multi-Source Training Data

Training data comprise nine Re-ID datasets selected to represent diverse acquisition conditions, including occlusion, clothing changes, and multi-camera video observations. Their respective image counts are summarized in Table 1. Together, the curated sources contain 782,064 images and 10,036 identities. The four public evaluation benchmarks are excluded from training and model selection.

3.2 Data Harmonization and Partitioning

The source datasets differ in directory structure, identity conventions, camera metadata, and annotation formats. Each source was audited and converted into a common metadata

Table 2. Query and gallery sizes of the unseen public evaluation benchmarks.

Dataset	Query	Gallery
Market-1501 [**?**]	3,368	19,732
MSMT17 [**?**]	11,659	82,161
DukeMTMC-reID [**?**]	2,228	17,661
Occluded-Duke [**?**]	2,210	17,661

schema with consistent sample identifiers, identity keys, and camera fields. Identity labels were remapped into a global index space containing 10,036 identities, with identity spaces kept separate across source datasets. Each source was partitioned into training and validation identities, using an approximately 80/20 split where applicable and respecting source-specific release protocols. Validation identities were excluded from optimization and reserved for checkpoint selection (Section 4.2).

The pooled training set is strongly imbalanced: LPW and RPIfield alone account for roughly two thirds of all images, whereas CUHK03-NP contributes less than one percent. Sampling identities uniformly from the pool would therefore let a few video-derived sources dominate the gradient, while sampling sources uniformly would repeatedly revisit the identities of the smallest datasets. We adopt an intermediate, domain-balanced scheme. Each minibatch is filled with P identity slots; for each slot, a source d is drawn with probability

$$\pi_d = \frac{\sqrt{N_d}}{\sum_{d'} \sqrt{N_{d'}}}, \tag{1}$$

where N_d is the number of training identities in source d. An identity is then taken from a per-source queue that is reshuffled whenever it is exhausted, so every identity of a source is visited before any is repeated, and identities already present in the current minibatch are skipped. Finally, K images of that identity are drawn without replacement from a per-identity pool, or with replacement when the identity has fewer than K images. With $P = 16$ and $K = 8$, every minibatch contains positives and negatives for each anchor, as required by batch-hard triplet mining, while the square-root weighting keeps large sources represented without allowing them to dominate.

3.3 Unseen Public Benchmarks

Cross-domain retrieval is evaluated on four public benchmarks excluded from training and model selection: Market-1501 [**?**], MSMT17 [**?**], DukeMTMC-reID [**?**], and Occluded-Duke [**?**]. These benchmarks cover different evaluation conditions, from conventional still-image retrieval to large-scale multi-camera and occlusion-heavy scenarios. Their query and gallery sizes are summarized in Table 2.

Occluded-Duke is derived from DukeMTMC; this relationship is acknowledged when interpreting results.

4 Methodology

4.1 Model Architecture

This work compares three existing convolutional architectures rather than introducing a new model. The primary eight-objective loss sweep uses the FastReID Strong Baseline (SBS) [**?**,**?**]. FastReID serves as the implementation framework, while the SBS combines a ResNet-50 backbone with instance-batch normalization (IBN) [**?**], generalized-mean (GeM) pooling, and a batch-normalization neck to produce a 2048-dimensional embedding. The cross-backbone comparison additionally includes OSNet [**?**] and ConvNeXt-S [**?**]. It compares the FastReID Circle-softmax-plus-triplet default against the candidate selected from the eight-configuration loss sweep. All architectures use the same multi-source training data, input resolution, optimization schedule, and evaluation protocols. Training is implemented in PyTorch using the FastReID codebase, extended with our multi-source data loader and domain-balanced sampler (Section 3.2).

4.2 Training Setup

The primary SBS loss sweep evaluates eight candidate configurations that differ in identity-learning objective and triplet supervision (Section 4.3). Each architecture is initialized using its corresponding ImageNet-pretrained weights. We use Adam with a base learning rate of 3.5×10^{-4} and weight decay of 5×10^{-4}. Cross-entropy-based heads use label smoothing with $\epsilon = 0.1$, while configurations incorporating triplet loss use hard mining with a zero margin.

Training proceeds for 50 *virtual epochs*, each comprising 2,000 parameter updates sampled from the multi-source loader. This definition accommodates the combined training pool, which is substantially larger than a single pass over all images. The backbone remains frozen for the first 1,000 updates and is subsequently optimized jointly with the embedding head. The learning rate follows cosine decay after a 20-epoch delay, with linear warmup over the first 1,000 iterations and a minimum learning rate of 7×10^{-7}. Minibatches contain 128 images, with eight instances per identity, using domain-balanced sampling as described in Section 3.2. Mixed-precision forward passes are used during training, whereas validation feature extraction is performed in full precision. Primary experiments are conducted using two GPU accelerators with data-parallel model replication. Each SBS objective is trained using three random seeds (0, 1, and 2), with all other hyperparameters held fixed. The separate cross-backbone comparison applies the same optimization and evaluation protocol to the FastReID Circle-softmax-plus-triplet default and the candidate selected from the primary loss sweep.

For checkpoint selection, we construct a composite validation protocol using source-specific identities, retaining official partitions where available and predefined cross-view or cross-context splits otherwise. The four public evaluation benchmarks are excluded from training, validation, and checkpoint selection. Every five virtual epochs, including the final epoch, we evaluate each validation protocol using L2-normalized cosine similarity. Gallery images sharing both identity and camera with a query are excluded. The selected checkpoint maximizes the unweighted mean validation score across protocols:

$$s_i = \frac{\text{Rank-1}_i + \text{mAP}_i}{2}. \tag{2}$$

The final checkpoint is selected according to the mean of s_i across validation protocols.

4.3 Identity-Learning Objectives in a Unified View

We compare different identity-learning heads using a common notation that distinguishes their target and non-target scores and the class structures they assume. The implementations follow the cited formulations; the objective configurations are summarized in Table 3.

Notation and cross-entropy heads. Let $\mathbf{f} \in \mathbb{R}^d$ denote the embedding of an image with identity label y, and let $C = 10{,}036$ denote the number of training identities. Linear softmax uses bias-free logits $z_j = \mathbf{w}_j^\top \mathbf{f}$, following the BNNeck design [**?**]. For angular heads, features and class centers are ℓ_2-normalized, such that

$$\cos\theta_j = \frac{\mathbf{w}_j^\top \mathbf{f}}{\|\mathbf{w}_j\|\|\mathbf{f}\|}, \tag{3}$$

where s denotes the logit scale. Cross-entropy-based heads use label smoothing with $\epsilon = 0.1$. The smoothed target distribution is defined as

$$q_j = \begin{cases} 1-\epsilon, & j = y, \\ \frac{\epsilon}{C-1}, & j \neq y. \end{cases} \tag{4}$$

Given the head-specific logits z_j and softmax probabilities p_j, the classification objective is

$$\mathcal{L}_{\text{CE}} = -\sum_{j=1}^{C} q_j \log p_j, \qquad p_j = \frac{\exp(z_j)}{\sum_{k=1}^{C}\exp(z_k)}. \tag{5}$$

SphereFace2 instead employs an independent binary-classification objective, described below.

Classification heads. Table 3 summarizes the compared heads. For Sub-center ArcFace, each identity is represented by K centers. The target identity uses its best-matching center, while each non-target identity is represented by its maximum-similarity center. The Circle-softmax formulation corresponds to the class-level variant implemented in the Strong Baseline rather than the pair-based Circle Loss formulation.

For Circle-softmax, the adaptive weights are defined as

$$\alpha_p = [1 + m - \cos\theta_y]_+, \qquad \alpha_n^j = [\cos\theta_j + m]_+, \tag{6}$$

with $\Delta_p = 1 - m$ and $\Delta_n = m$. These weights are treated as constants during backpropagation. This is the class-level head used by FastReID, rather than the pair-based objective introduced in the original Circle Loss formulation [**?**,**?**].

Table 3. Unified view of the compared identity-learning heads. θ_j denotes the angle between a feature and a class center, while z_y and z_j denote target and non-target logits for cross-entropy heads.

Head	Target score	Non-target score	Class assumption	Settings
Linear softmax [**?**]	$\mathbf{w}_y^\top \mathbf{f}$	$\mathbf{w}_j^\top \mathbf{f}$	One unnormalized vector per identity	None
ArcFace [**?**]	$s\cos(\theta_y + m)$	$s\cos\theta_j$	One center; additive angular margin	$s = 64$, $m = 0.35$
Sub-center ArcFace [**?**]	$s\cos(\theta_{y,k^*} + m)$	$s\max_k \cos\theta_{j,k}$	K centers per identity	$K = 3$
SphereFace2 [**?**]	$r(g(\cos\theta_y) - m) + b$	$r(g(\cos\theta_j) + m) + b$	Independent per-class binary decisions	$r = 40$, $m = 0.4$
Circle-softmax [**?**,**?**]	$s\alpha_p(\cos\theta_y - \Delta_p)$	$s\alpha_n^j(\cos\theta_j - \Delta_n)$	One center; adaptive per-logit weights	$s = 64$, $m = 0.35$

SphereFace2 replaces the softmax over identities with C independent binary classifiers [**?**]. We use its SphereFace2-C formulation with an additive margin and similarity adjustment:

$$g(z) = 2\left(\frac{z+1}{2}\right)^t - 1. \tag{7}$$

The resulting objective is

$$\begin{aligned}\mathcal{L}_{\mathrm{SF2}} = {} & \frac{\omega\lambda}{r}\log\left(1 + \exp\left[-\left(r(g(\cos\theta_y) - m) + b\right)\right]\right) \\ & + \frac{\omega(1-\lambda)}{r}\sum_{j\neq y}\log\left(1 + \exp\left[r(g(\cos\theta_j) + m) + b\right]\right).\end{aligned} \tag{8}$$

Here, r denotes the hypersphere radius, m the margin, λ balances the positive term against the negative terms, b is a learnable bias, and ω is a constant loss weight. We adopt the reference implementation settings $r = 40$, $m = 0.4$, $\lambda = 0.9$, $t = 3$, and $\omega = 50$. The bias b is initialized to balance the positive and negative terms at the beginning of training. As in the reference implementation, the similarity adjustment and margin are treated as constant offsets during backpropagation, with gradients computed with respect to the cosine similarities.

Triplet loss and evaluated configurations. When enabled, hard-mined triplet loss is added to the head-specific objective. The configured margin is zero, corresponding to the soft-margin formulation used in the FastReID implementation:

$$\mathcal{L}_{\mathrm{tri}} = \log\left(1 + \exp(d_{ap} - d_{an})\right), \tag{9}$$

where d_{ap} and d_{an} denote the Euclidean distances from an anchor to its hardest positive and hardest negative, respectively. These distances are computed using the unnormalized BNNeck features. Mining is performed within each device's sub-batch of 64 images, containing eight identities with eight images per identity.

Table 4. Eight candidate objective configurations evaluated using the shared SBS R50-IBN architecture. Each head is trained with and without triplet loss.

Classifier / head	Losses	Reference
Linear softmax	CE	[**?**]
ArcFace ($s = 64, m = 0.35$)	CE	[**?**]
Sub-center ArcFace ($K = 3$)	CE	[**?**]
Angular head	SphereFace2	[**?**]
Linear softmax	CE + Triplet	[**?**,**?**]
ArcFace	CE + Triplet	[**?**,**?**]
Sub-center ArcFace ($K = 3$)	CE + Triplet	[**?**,**?**]
Angular head	SphereFace2 + Triplet	[**?**,**?**]

The primary loss sweep comprises eight candidate configurations: linear softmax, ArcFace, Sub-center ArcFace, and SphereFace2, each evaluated both without and with hard-mined triplet loss. Thus, triplet supervision is an explicit comparison factor for every candidate head. The FastReID Circle-softmax-plus-triplet default is not part of this sweep; it is reserved as the reference objective in the separate comparison across the SBS, OSNet, and ConvNeXt-S. All eight candidate configurations use the same FastReID Strong Baseline architecture; only the identity-learning head and the use of triplet loss vary.

4.4 Evaluation Metrics

We evaluate retrieval performance using mean average precision (mAP) and the cumulative matching characteristic (CMC), following the standard query and gallery protocols of each benchmark. CMC performance is summarized using Rank-1 accuracy, the proportion of queries for which a correct match appears at the top of the ranked gallery. Full CMC curves up to Rank-20 are reported for selected models. Extracted features are ℓ_2-normalized and ranked using cosine similarity. No re-ranking is applied. All models are evaluated using identical query and gallery roles within each benchmark, ensuring consistent comparisons across objective configurations and architectures.

5 Results

The primary objective comparison comprises 24 training runs (eight candidate configurations × three seeds). Held-out retrieval is reported on four public benchmarks Market-1501, MSMT17, DukeMTMC-reID, and Occluded-Duke using the unseen-target evaluation bundle. Each table cell gives mean mAP / Rank-1 (%) with seed standard deviations (±). The average column is the unweighted mean across the four benchmarks. One Sub-center ArcFace run (seed 2 without triplet) diverged during training and reached a macro Rank-1 below 2%; it is treated as a failed run and excluded, so that configuration is summarized over seeds 0 and 1.

Table 5. Objective comparison without triplet loss: mean mAP / Rank-1 (%) over seeds 0–2 on four held-out benchmarks. Sub-center ArcFace is averaged over seeds 0–1 (seed 2 diverged).

Objective	Market	MSMT17	Duke	Occ-Duke	Avg.
Linear softmax	58.7±0.5 / 81.0±0.6	16.2±0.3 / 36.3±0.5	48.8±0.4 / 65.7±0.7	21.0±0.7 / 26.9±0.6	36.2±0.2 / 52.5±0.2
ArcFace	69.8±0.3 / 87.8±0.2	23.5±0.4 / 49.3±0.5	56.9±0.5 / 73.4±0.6	28.9±0.5 / 37.1±0.6	44.8±0.3 / 61.9±0.1
Sub-center ArcFace	64.1±0.2 / 85.6±0.2	20.8±0.3 / 45.6±0.2	52.7±0.2 / 69.7±0.6	25.8±0.1 / 33.9±0.2	40.8±0.1 / 58.7±0.3
SphereFace2	60.3±1.4 / 83.5±0.4	16.7±1.0 / 37.9±1.2	43.6±4.3 / 65.6±2.5	21.0±2.3 / 31.9±2.2	35.4±2.2 / 54.7±1.4

Table 6. Objective comparison with triplet loss: mean mAP / Rank-1 (%) over seeds 0–2 on four held-out benchmarks.

Objective	Market	MSMT17	Duke	Occ-Duke	Avg.
Linear softmax	53.6±1.3 / 77.2±0.9	15.1±0.7 / 34.7±1.0	46.8±0.7 / 65.1±1.3	20.0±0.3 / 26.7±0.6	33.9±0.2 / 50.9±0.4
ArcFace	70.4±0.8 / 87.9±0.4	22.0±0.2 / 46.6±0.4	56.3±0.5 / 72.4±0.6	28.7±0.5 / 35.8±0.5	44.3±0.2 / 60.7±0.1
Sub-center ArcFace	67.6±0.6 / 87.1±0.4	20.7±0.4 / 44.4±0.8	54.1±0.5 / 70.3±0.6	26.2±0.8 / 33.3±1.1	42.1±0.5 / 58.8±0.7
SphereFace2	62.2±2.0 / 84.4±0.9	16.3±0.8 / 37.8±1.9	45.2±0.7 / 66.4±0.6	23.0±0.8 / 34.9±0.4	36.7±0.7 / 55.9±0.7

5.1 Objective Comparison

Tables 5 and 6 compare the evaluated classification objectives without and with triplet supervision, respectively. This comparison examines their retrieval performance under the same SBS architecture and multi-source training protocol.

ArcFace without triplet achieves the highest four-benchmark average of all eight candidates (44.8% mAP and 61.9% Rank-1), with low variation across seeds (±0.3 mAP and ±0.1 Rank-1). Its margin is not driven by a single dataset: it is the best or within one standard deviation of the best configuration on every benchmark, and its advantage is largest on MSMT17, the largest and most heterogeneous target, where it reaches 23.5% mAP and 49.3% Rank-1. The ranking of heads is the same with and without triplet loss: ArcFace is followed by Sub-center ArcFace, while SphereFace2 and linear softmax trail by 8–10 points in average mAP.

The two head families with an angular margin outperform the unnormalized linear softmax by a wide margin, which is consistent with the view that normalizing features and class centers produces an embedding geometry that ranks better under the cosine similarity used at retrieval time. Representing each identity with three sub-centers does not help on average, however. Sub-center ArcFace was designed to absorb label noise and intra-class multimodality, both of which are present in our pooled sources, yet it remains 4.0 points below single-center ArcFace in average mAP. One plausible explanation is that sub-centers fit source-specific appearance modes, such as different clothing or scenes of the same identity, that the retrieval embedding then fails to merge at test time.

Figure 2 isolates the effect of triplet supervision. Its sign depends on the head. Triplet loss lowers average mAP by 2.3 points for linear softmax and by 0.5 points for ArcFace, but raises it by 1.3 points for both Sub-center ArcFace and SphereFace2. It also reduces the seed variance of SphereFace2 from 2.2 to 0.7 points, which suggests that the metric term mainly stabilizes the weaker heads rather than adding complementary information to a well-structured angular embedding. For ArcFace, the small drop in average mAP hides a larger shift on MSMT17 (-1.5 mAP and -2.7 Rank-1), whereas Market-1501 changes only slightly in the opposite direction. Triplet supervision is therefore not a free

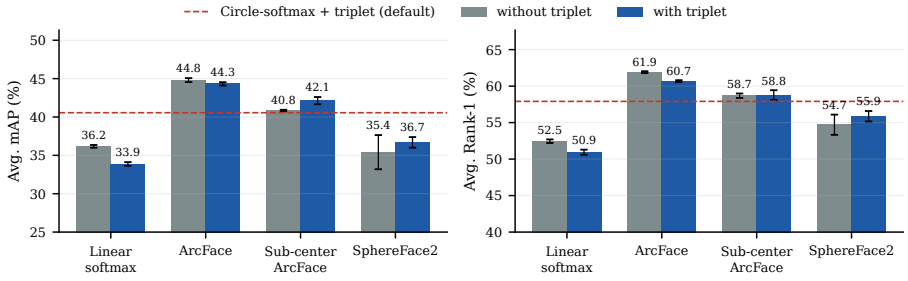


Fig. 2. Effect of triplet supervision for each identity-learning head on the SBS R50-IBN (average over the four held-out benchmarks; error bars show the standard deviation across seeds). The dashed line marks the FastReID Circle-softmax-plus-triplet default.

Table 7. Cross-backbone comparison (mAP / Rank-1, %; mean±std over seeds 0–2 on four held-out benchmarks).

Backbone	Objective	Market	MSMT17	Duke	Occ-Duke	Avg.
SBS R50-IBN	Circle-softmax + Triplet	67.4±1.3 / 86.7±0.6	15.5±0.2 / 37.5±0.4	52.5±1.1 / 71.2±1.1	26.9±1.0 / 36.2±1.2	40.6±0.8 / 57.9±0.6
SBS R50-IBN	ArcFace	69.8±0.3 / 87.8±0.2	23.5±0.4 / 49.3±0.5	56.9±0.5 / 73.4±0.6	28.9±0.5 / 37.1±0.6	44.8±0.3 / 61.9±0.1
OSNet	Circle-softmax + Triplet	69.3±0.7 / 87.3±0.4	21.5±1.1 / 48.4±1.5	56.3±0.2 / 72.7±0.2	27.3±0.4 / 34.8±0.3	43.6±0.2 / 60.8±0.4
OSNet	ArcFace	64.7±0.4 / 85.9±0.3	23.0±0.2 / 50.9±0.5	54.4±0.1 / 72.5±0.4	25.9±0.7 / 34.8±1.1	42.0±0.1 / 61.0±0.2
ConvNeXt-S	Circle-softmax + Triplet	59.0±1.4 / 81.3±0.9	13.7±0.8 / 33.4±1.3	51.5±1.8 / 70.0±1.2	25.2±1.5 / 35.0±2.2	37.4±1.3 / 54.9±1.3
ConvNeXt-S	ArcFace	61.9±0.2 / 83.9±0.2	15.3±0.6 / 36.3±1.3	55.1±0.4 / 72.2±0.4	27.7±0.5 / 36.8±0.7	40.0±0.3 / 57.3±0.5

addition, and its inclusion in default recipes deserves re-examination when the identity head changes.

Finally, all ArcFace and Sub-center ArcFace variants exceed the FastReID Circle-softmax-plus-triplet default on the same architecture (40.6% mAP and 57.9% Rank-1; Table 7). Since ArcFace with triplet and the default differ only in the classification head under our shared protocol, the 3.7-point gap between them can be attributed to the head itself.

5.2 Cross-Backbone Comparison

To examine whether the best objective of the sweep transfers beyond the primary architecture, we train ArcFace without triplet on OSNet and ConvNeXt-S with the same data, schedule, and seeds, and compare it with the FastReID Circle-softmax-plus-triplet default on each backbone. Table 7 summarizes retrieval on the same four held-out benchmarks.

With the FastReID default objective, the ranking of backbones is inversely related to their size. OSNet, with fewer than 2M deployed parameters, achieves the strongest four-benchmark average (43.6% mAP and 60.8% Rank-1), 3.0 mAP points above the Strong Baseline, and the gain is concentrated on MSMT17 (+6.0 mAP and +10.9 Rank-1). ConvNeXt-S, the largest model, is the weakest (37.4% and 54.9%) and also the least stable across seeds. Its larger capacity may be harder to exploit with the available

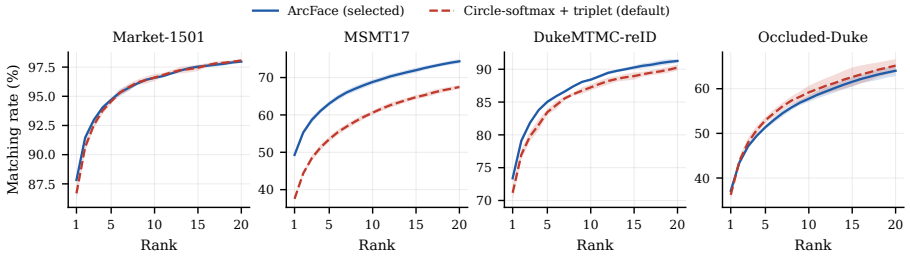


Fig. 3. CMC curves (Ranks 1–20) of ArcFace without triplet and the FastReID Circle-softmax-plus-triplet default on SBS R50-IBN. Lines show the mean over seeds 0–2 and shaded bands one standard deviation. Note the different y-axis ranges.

training data and a schedule tuned for ResNet-style networks, but this comparison does not isolate model size, data scale, or architecture-specific tuning as causes.

Changing the objective does not move the three backbones in the same direction. On the Strong Baseline, ArcFace raises the average by 4.2 mAP and 4.0 Rank-1, to 44.8% and 61.9%, which is the best result in the study and exceeds every OSNet run. On ConvNeXt-S the same swap helps on all four benchmarks, by 2.6 mAP and 2.4 Rank-1 on average, and it narrows but does not close the gap to the Strong Baseline. On OSNet it does the opposite: average mAP falls by 1.6 points, from 43.6% to 42.0%, while Rank-1 is essentially unchanged (60.8% to 61.0%). The OSNet loss is concentrated where the default was already strong, with −4.6 mAP on Market-1501, −1.9 on DukeMTMC-reID, and −1.4 on Occluded-Duke. MSMT17 is the only benchmark on which ArcFace improves all three architectures, by 8.0, 1.5, and 1.6 mAP respectively.

The OSNet regression is unlikely to be noise: the three seeds agree to ±0.1 mAP. A plausible cause is the margin itself. We used the face-recognition reference settings, a scale of 64 and a margin of 0.35, on every backbone, and OSNet's embedding is 512-dimensional rather than 2048. A margin tuned for a large embedding can over-separate a compact one and discard the within-identity variation that retrieval on the easier benchmarks rewards. We did not retune the margin, so this remains an explanation rather than a demonstrated cause. What the experiment does show is that the best objective and the best architecture are not independent choices. Selecting ArcFace on the Strong Baseline and applying it unchanged made OSNet worse, and the strongest deployed system is the Strong Baseline with ArcFace, not the smallest model.

5.3 Retrieval Performance Analysis

Figure 3 compares the CMC curves of ArcFace and the FastReID default on the Strong Baseline up to Rank-20.

The curves show that the benefit of ArcFace is far from uniform. On MSMT17, ArcFace dominates at every rank, with an 11.8-point lead at Rank-1 that narrows only to about 7 points at Rank-20; the default objective does not merely misorder the top of the list but fails to retrieve many correct matches at all. On DukeMTMC-reID, ArcFace

Table 8. Model size and inference cost at an input resolution of 384×128, measured on the authors' laptop (Lenovo Legion 5; NVIDIA GeForce RTX 5070 Laptop GPU, 8 GB; Intel Core Ultra 9 275HX, 24 cores; 32 GB RAM; AC power). Deployed parameters exclude the identity classifier, which is used only during training. MACs are counted for a single image. Latency is the time to embed one image (batch size 1, FP32); GPU throughput uses FP16 autocast at batch size 64. All timings are medians over three independent repeats.

	Params (M)			GPU		CPU	Embed.
Model	Deployed	Training	GMACs	Latency (ms)	Throughput (img/s)	Latency (ms)	dim.
SBS R50-IBN	23.5	44.1	6.09	8.9	1002	28.6	2048
OSNet	1.9	7.0	1.47	12.7	1462	24.8	512
ConvNeXt-S	49.5	57.2	10.25	11.0	657	35.1	768

keeps a consistent lead of 1–2 points at every rank. On Market-1501, ArcFace leads by 1.1 points at Rank-1, but the two curves are indistinguishable from Rank-5 onward, which suggests that this benchmark is close to saturation for both objectives. Occluded-Duke shows the opposite trend: ArcFace leads by 0.9 points at Rank-1, but the default overtakes it from Rank-2 and stays 1.0–1.5 points ahead up to Rank-20. Under heavy occlusion, the triplet term in the default objective may therefore help recover partially visible matches at deeper ranks, even though it does not improve the first match. Taken together, ArcFace yields the largest gains on the most heterogeneous target and offers no advantage at deeper ranks under occlusion.

5.4 Computational Cost

Table 8 compares the size and inference cost of the three architectures, and Fig. 4 relates them to retrieval accuracy. We distinguish two parameter counts. During training, each model carries an identity classifier with one center per training identity (10,036 rows), which for the Strong Baseline amounts to almost half of all weights. This classifier is discarded at retrieval time, so only the backbone, pooling, and neck are deployed. Since every classifier is discarded at retrieval time, the deployed cost of a backbone is identical for all of its objectives. Switching from the default objective to ArcFace is therefore free at inference.

Measurement setup. All timings were obtained on a single consumer laptop, a Lenovo Legion 5 with an NVIDIA GeForce RTX 5070 Laptop GPU (8 GB, driver 577.05), an Intel Core Ultra 9 275HX CPU (24 cores, all used for CPU inference), and 32 GB of RAM, running Windows 11 on AC power. We used PyTorch 2.12 with CUDA 12.8 and cuDNN 9.2 in eager mode, without TensorRT, ONNX export, quantization, or operator fusion; the numbers therefore describe an unoptimized research deployment. MACs are counted with the PyTorch FLOP counter, where one multiply-accumulate corresponds to two floating-point operations. Each model and device is timed in a separate process after a 90-second cool-down, so that heat from one measurement does not throttle the next. Within each run, timing starts after warm-up iterations; we report the median of the per-repeat medians over three repeats, which is robust to occasional fast or throttled repeats. Absolute speeds depend on the hardware, power, and thermal configuration of a laptop and should mainly be read as relative comparisons between architectures.

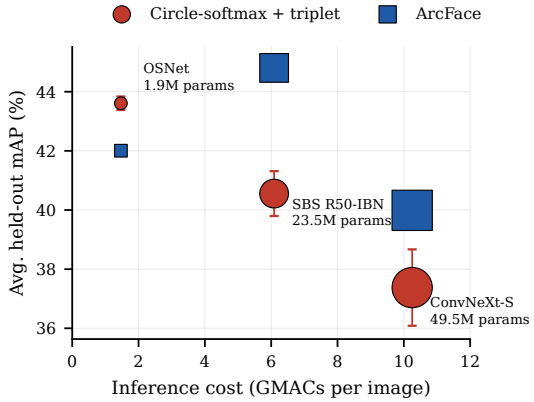


Fig. 4. Average held-out mAP versus inference cost. Circles use the FastReID default objective and squares use ArcFace; the two markers of a backbone share its compute cost. Marker area is proportional to the number of deployed parameters, and error bars show the standard deviation across seeds.

Efficiency comparison. OSNet is the most efficient model in terms of size and batched throughput: compared with the Strong Baseline, it has 12× fewer deployed parameters and 4× fewer MACs, processes about 1.5× more images per second on the GPU, and produces a 512-dimensional instead of a 2048-dimensional embedding, which reduces gallery storage and matching cost by a factor of four. With the default objective, it is also more accurate. ConvNeXt-S, in contrast, requires twice the parameters and 1.7× the MACs of the Strong Baseline and reaches only two thirds of its GPU throughput, yet it is the least accurate of the three.

The single-image latencies show that MACs are a poor proxy for speed in this regime. On the GPU, all three models take 9–13 ms per image regardless of their compute, because at batch size 1 the runtime is dominated by the overhead of launching many small kernels rather than by arithmetic. OSNet, with the deepest stack of small layers, is in fact the slowest of the three despite having the fewest MACs. On the CPU, OSNet is the fastest model (24.8 ms), but it is only 13% faster than the Strong Baseline despite its 4× lower MAC count, since its depthwise and pointwise convolutions are memory-bound and use the CPU's vector units less efficiently than dense convolutions. ConvNeXt-S is the slowest on the CPU (35.1 ms). All three models nevertheless embed a single person crop in under 40 ms on a laptop CPU alone, which is sufficient for many real-time tracking pipelines without a GPU. Finally, the accuracy gain from changing the objective is free at inference but not guaranteed. On the Strong Baseline, ArcFace adds 4.2 mAP at no extra cost, more than the gain from switching to OSNet. On OSNet the same change removes 1.6 mAP, so the deployed objective should be chosen per architecture.

6 Discussion

The objective matters as much as the backbone. A common practice in Re-ID is to inherit the loss recipe of a strong baseline and spend the remaining effort on the architecture. Our results suggest that this ordering is not justified under multi-source training. Replacing the Circle-softmax-plus-triplet default with plain ArcFace improves the Strong Baseline by 4.2 points in average mAP, which is more than the 3.0 points gained by switching to OSNet and is obtained without changing the deployed model. The gain is largest on MSMT17, the target with the most cameras, scenes, and identities, which indicates that the margin-based angular embedding mainly helps when the domain gap is large rather than refining already strong rankings.

The best loss on one backbone is not a default for another. The cross-backbone experiment asks whether the winning objective is a property of the training data or of the architecture it was selected on. It is partly the latter. ArcFace improves the Strong Baseline and ConvNeXt-S but reduces the average mAP of OSNet, which was the most accurate model under the default loss. The one result that does transfer is the MSMT17 gain, which is positive for all three architectures, although it is much larger for the Strong Baseline. A practical consequence is that a published loss recipe should be re-checked on the architecture that will actually be deployed. The best raw accuracy in this study comes from the Strong Baseline with ArcFace. The best accuracy per parameter remains OSNet with the default objective, which trails that result by 1.2 mAP while using 12× fewer deployed parameters.

Triplet loss is not a universal complement. The common recipe of pairing a classification loss with batch-hard triplet loss does not hold uniformly in our setting. The triplet term helps the two heads that are weakest or least stable on their own, Sub-center ArcFace and SphereFace2, but slightly hurts ArcFace and clearly hurts linear softmax. A likely reason is that, with a large and well-separated angular margin, the classification loss already enforces the relative ordering that the triplet term targets, while the hardest pairs mined from a heterogeneous multi-source batch are dominated by source-specific nuisances such as scene and resolution. The CMC analysis adds a caveat: under heavy occlusion, the default objective with triplet loss retrieves more correct matches at deeper ranks, so the metric term may still be useful for occlusion-heavy deployments.

Open-set formulations and multi-center heads. SphereFace2, motivated by open-set recognition, does not outperform ArcFace in this study. Training uses closed-set identity labels from the source data, whereas evaluation ranks embeddings on held-out domains after discarding the classifier; its open-set motivation therefore does not by itself predict retrieval performance. It was also the least stable head across seeds. Likewise, Sub-center ArcFace, which is designed to tolerate label noise and multimodal identities, did not benefit from the heterogeneous labels of our pooled sources. These results do not establish that either head is generally unsuitable, since we used the reference hyperparameters without tuning them for Re-ID.

Capacity and efficiency. With the default objective, accuracy decreases as model size increases: OSNet is both the smallest and the most accurate backbone, and ConvNeXt-S is both the largest and the least accurate. The weaker ConvNeXt-S result may reflect a mismatch between model capacity and the effective scale or diversity of the training data, or a schedule that was not tuned for this architecture; these explanations were not tested directly. From a deployment perspective the choice depends on the budget. Where accuracy matters most, the Strong Baseline with ArcFace is the best system we trained. Where parameters, gallery size, or CPU throughput dominate, OSNet with the default objective gives up 1.2 mAP and keeps a 512-dimensional embedding. Throughout, differences are interpreted descriptively alongside the variation across three seeds, without formal significance tests.

7 Conclusion

This study asked whether the identity-learning objective changes how well a Re-ID model trained on nine heterogeneous sources generalizes to unseen camera networks. On the FastReID Strong Baseline, the answer is yes. Across eight configurations and three seeds, ArcFace without triplet loss gave the best average retrieval on four held-out benchmarks, 4.2 mAP and 4.0 Rank-1 above the FastReID default, with the largest gain on MSMT17. Triplet supervision helped only the weaker heads, so it should be chosen for the head it is paired with rather than added by default. The answer is also that this improvement is not a property of the objective alone. The same ArcFace settings raise ConvNeXt-S by 2.6 mAP and lower OSNet by 1.6, and the best system in the study is the Strong Baseline with ArcFace rather than the smallest or the largest backbone. Because the classifier is discarded at retrieval, that choice costs nothing at inference, but it has to be confirmed on the architecture being deployed. The compact OSNet with the default objective remains the strongest accuracy-per-parameter option. Completing the cross-source duplicate review, retuning the angular margin per backbone, and repeating the comparison on Transformer architectures are the natural next steps.

8 Limitations

The eight-objective sweep was run only on the Strong Baseline, under a fixed fifty-epoch budget and a single optimization schedule. OSNet and ConvNeXt-S were compared on two objectives, the FastReID default and the ArcFace configuration that won the sweep, so we cannot claim that ArcFace is the best head for those architectures, only that it is better or worse than the default. Every head used its reference hyperparameters. In particular, the ArcFace scale and margin were not retuned for OSNet's 512-dimensional embedding, and the drop in OSNet accuracy may narrow or reverse with a smaller margin. The shared schedule was also not adapted to ConvNeXt-S, whose weaker results may partly reflect optimization rather than capacity.

ArcFace was selected because it performed best on the same held-out benchmarks used for reporting, so its Strong Baseline result may be slightly optimistic. The margin over the default, 4.2 mAP, is far larger than the variation across seeds. The OSNet and ConvNeXt-S models were trained only after this choice, and the OSNet regression was

not a result we selected for. One Sub-center ArcFace run diverged and was excluded, leaving two seeds for that configuration. With three seeds and no formal significance test, small differences, such as the 0.5 mAP gap between ArcFace with and without triplet loss, should be read as consistent trends rather than as precise effect sizes. Checkpoints were chosen by a pooled identity-holdout validation, not by the public benchmarks, so the selected epoch need not be the best one on the reported test sets.

The cross-source pixel-level duplicate check across the nine training sources and the test sets is not yet complete, so strict acquisition independence cannot be guaranteed. DukeMTMC-reID and Occluded-Duke share their acquisition, and the two benchmarks should not be treated as independent evidence; the availability of DukeMTMC also depends on licensing. Evaluation covers still-image retrieval only. We report no video benchmark, no Transformer backbone, and no controlled ablation of label noise, identity multimodality, model capacity, or data scale, so our explanations of those effects remain qualitative. Inference costs were measured on one consumer laptop in eager-mode PyTorch and may rank differently on server or embedded hardware or with an optimized runtime.

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